

CHAPTER – 1

ACCIDENTS IN INDIA

Introduction

This is the eighth edition of the report after the comprehensive revision of the proforma for data collection of 'Accidental Deaths & Suicides in India' was done in 2014 in consultation with States/UTs Police and others stakeholders.

An accident is an unplanned event that results in injury, deaths, damage to property or some other losses. For statistical purpose, Accidental Deaths have been classified as deaths that could be due to forces/factors of the nature which have been termed as

'Accidental Deaths due to Forces of Nature' or could be due to deliberate or negligent conduct of human beings, which is termed as 'Other causes of Accidental Deaths'.

The rate of accidental deaths (per lakh of population) has increased in 2021 as compared to 2020. The total number of accidental deaths in 2021 is 3,97,530. [LIST-1.1]. The incidence and rate of accidental deaths during 2011 – 2021 is presented in Table-1.1.

LIST-1.1
Percentage Change in Population, Number and Rate of Accidental Deaths over the Corresponding Previous Year during 2017 to 2021

Year	Mid-Year Projected Population (In Lakh ⁺)	Percentage Change in Population over the Previous Year	Total Number of Accidental Deaths	Percentage Change in Accidental Deaths over the Previous Year	Rate of Accidental Deaths	Percentage Change in Rate of Accidental Deaths over the Previous Year
(1)	(2)	(3)	(4)	(5)	(6)	(7)
2017	13091.6 [#]	2.8	396584	-5.2	30.3	-7.6
2018	13233.8 [#]	1.1	411824	3.8	31.1	2.6
2019	13376.1 [#]	1.1	421104	2.3	31.5	1.3
2020	13533.9 ^{\$}	1.2	374397	-11.1	27.7	-12.1
2021	13671.8 ^{\$}	1.0	397530	6.2	29.1	5.1

Source: Report of the Technical Group on Population Projections (November, 2019), National Commission on Population, Ministry of Health & Family Welfare.

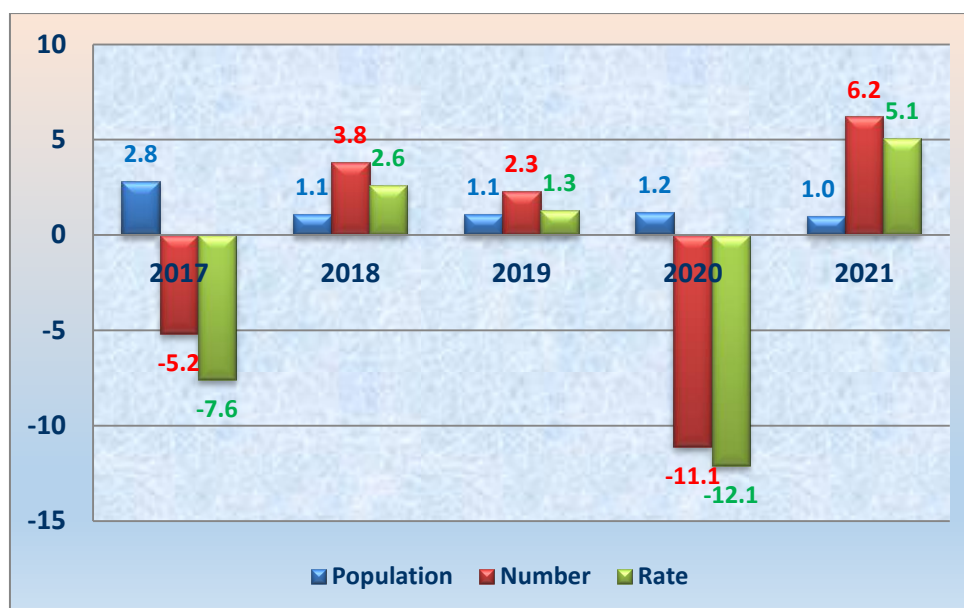
\$ Source: Report of the Technical Group on Population Projections (July, 2020), National Commission on Population, Ministry of Health & Family Welfare

+ One Lakh = 0.1 Million = 100,000

As per data provided by States/UTs

FIGURE-1.1

Percentage Change in Population, Number and Rate of Accidental Deaths during 2017 – 2021 (Over Corresponding Previous Year)



- As per data provided by States/UTs.

Out of 3,97,530 accidental deaths, 7,126 (1.8%) deaths were due to forces of nature and 3,90,404 (98.2%) deaths were due to other causes.

There was a decrease of 3.8% (from 7,405 deaths to 7,126 deaths) in deaths due to forces of nature and an increase of 6.4% (from 3,66,992 deaths to 3,90,404 deaths) in deaths due to other causes during 2021 as compared to 2020 **LIST-1.2 and Table 1.3.**

LIST-1.2

Number of Accidental Deaths due to Forces of Nature & Other Causes (2017 to 2021)

Year	Forces of Nature		Other Causes		Total Accidental Deaths	
	Number of Deaths	% change over previous year	Number of Deaths	% change over previous year	Number of Deaths	% change over previous year
(1)	(2)	(3)	(4)	(5)	(6)	(7)
2017	7,143	-17.7	3,89,441	-4.9	3,96,584	-5.2
2018	6,891	-3.5	4,04,933	4.0	4,11,824	3.8
2019	8,145	18.2	4,12,959	2.0	4,21,104	2.3
2020	7,405	-9.1	3,66,992	-11.1	3,74,397	-11.1
2021	7,126	-3.8	3,90,404	6.4	3,97,530	6.2

- As per data provided by States/UTs.

FIGURE-1.2
STATE/UT – WISE ACCIDENTAL DEATHS DURING 2021



- As per data provided by States/UTs.

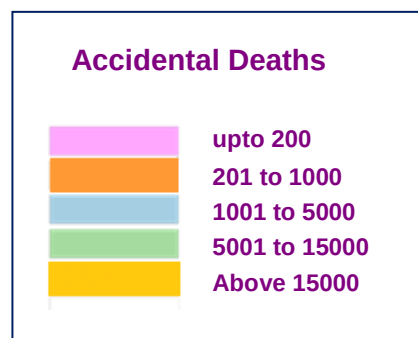
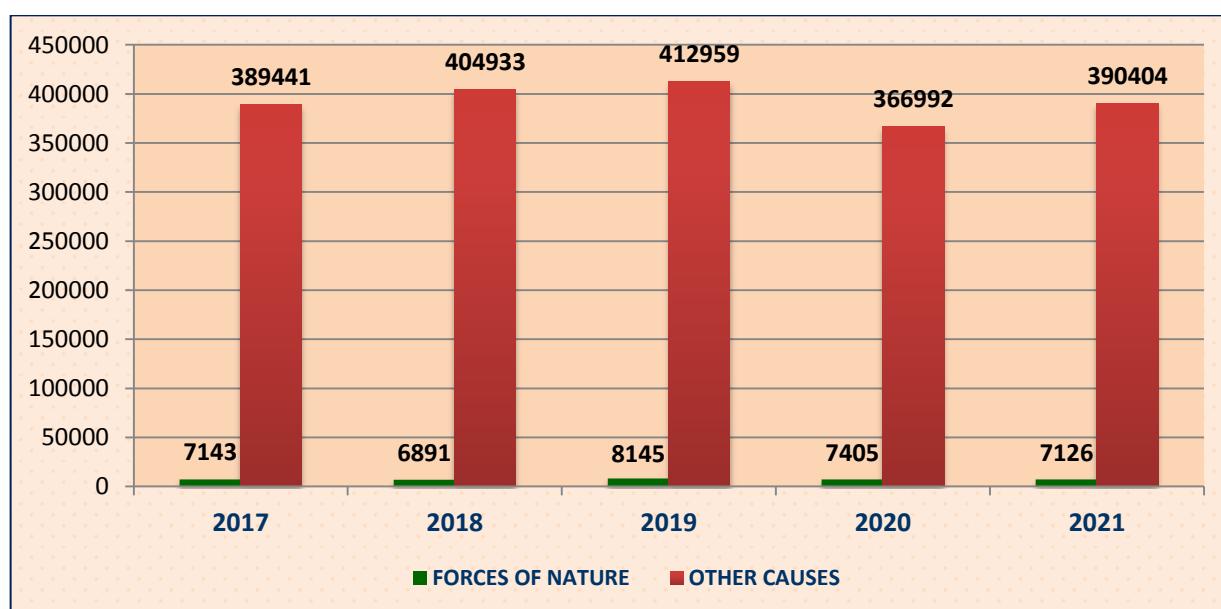


FIGURE-1.3

Total Number of Accidental Deaths due to Forces of Nature and Other Causes during 2017-2021



- As per data provided by States/UTs.

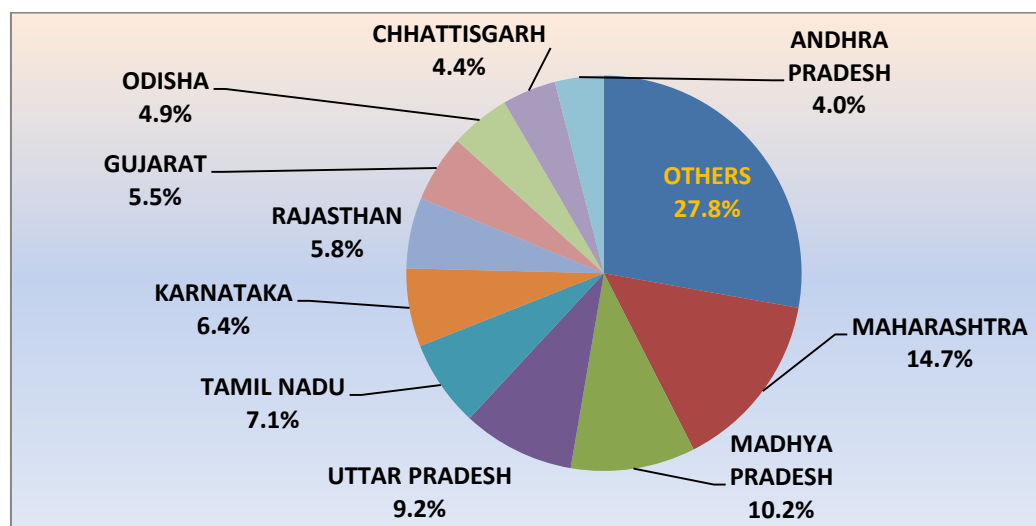
Number & Rate of Accidental Deaths in States/UTs

Maharashtra with a population share of 9.1% has reported highest number of accidental deaths (58,242), contributing nearly one-seventh (14.7%) of total accidental deaths reported in the country. Uttar Pradesh, the most

populous State in the country with population share of 16.9% has accounted for (36,521) 9.2% of the total accidental deaths in the country. The other States having higher percentage share in total accidental deaths were Madhya Pradesh (10.2%), Tamil Nadu (7.1%), Karnataka (6.4%), Rajasthan (5.8%), Gujarat (5.5%), Odisha (4.9%), Chhattisgarh (4.4%) and Andhra Pradesh (4.0%) [Table-1.2].

FIGURE-1.4

Percentage Share of Major States in Accidental Deaths during 2021



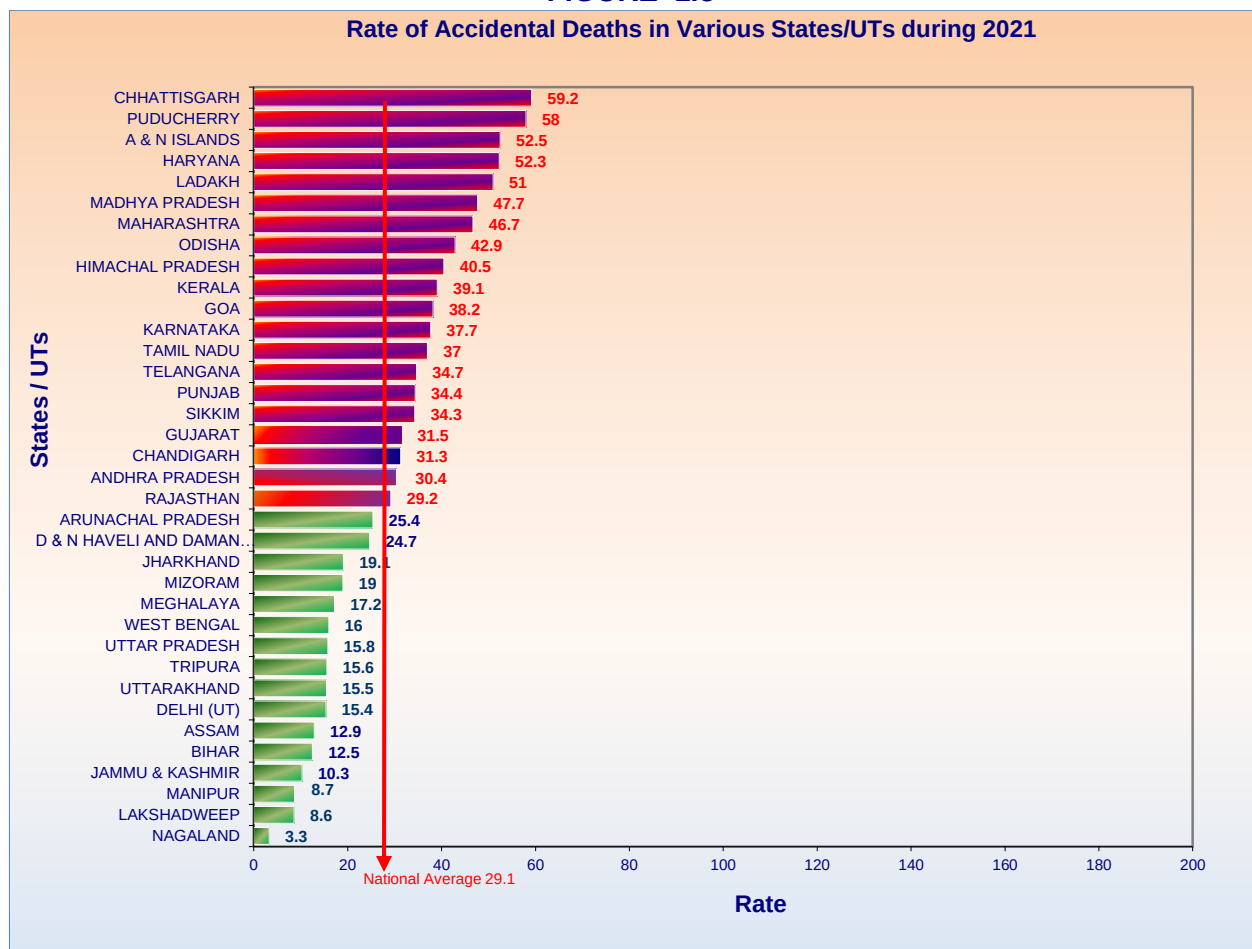
- As per data provided by States/UTs.

Other States/UTs include, Arunachal Pradesh, Assam, Bihar, Goa, Haryana, Himachal Pradesh, Jharkhand, Kerala, Manipur, Mizoram, Meghalaya, Nagaland, Punjab, Sikkim, Telangana, Tripura, Uttarakhand, West Bengal, Chandigarh, A & N Island, D&N Haveli and Daman & Diu, Delhi, Jammu & Kashmir, Ladakh, Lakshadweep and Puducherry

The highest **rate of accidental deaths** was reported from Chhattisgarh (59.2) followed by Puducherry (58), A&N Islands (52.5), Haryana (52.3), Ladakh (51), Madhya Pradesh (47.7) and Maharashtra (46.7) against the national average of 29.1. Twenty (20) out of 36

States/UTs have reported higher rates of 'Accidental Deaths' as compared to All India Average of 29.1 deaths per one lakh of population [**Figure – 1.5**].

FIGURE–1.5



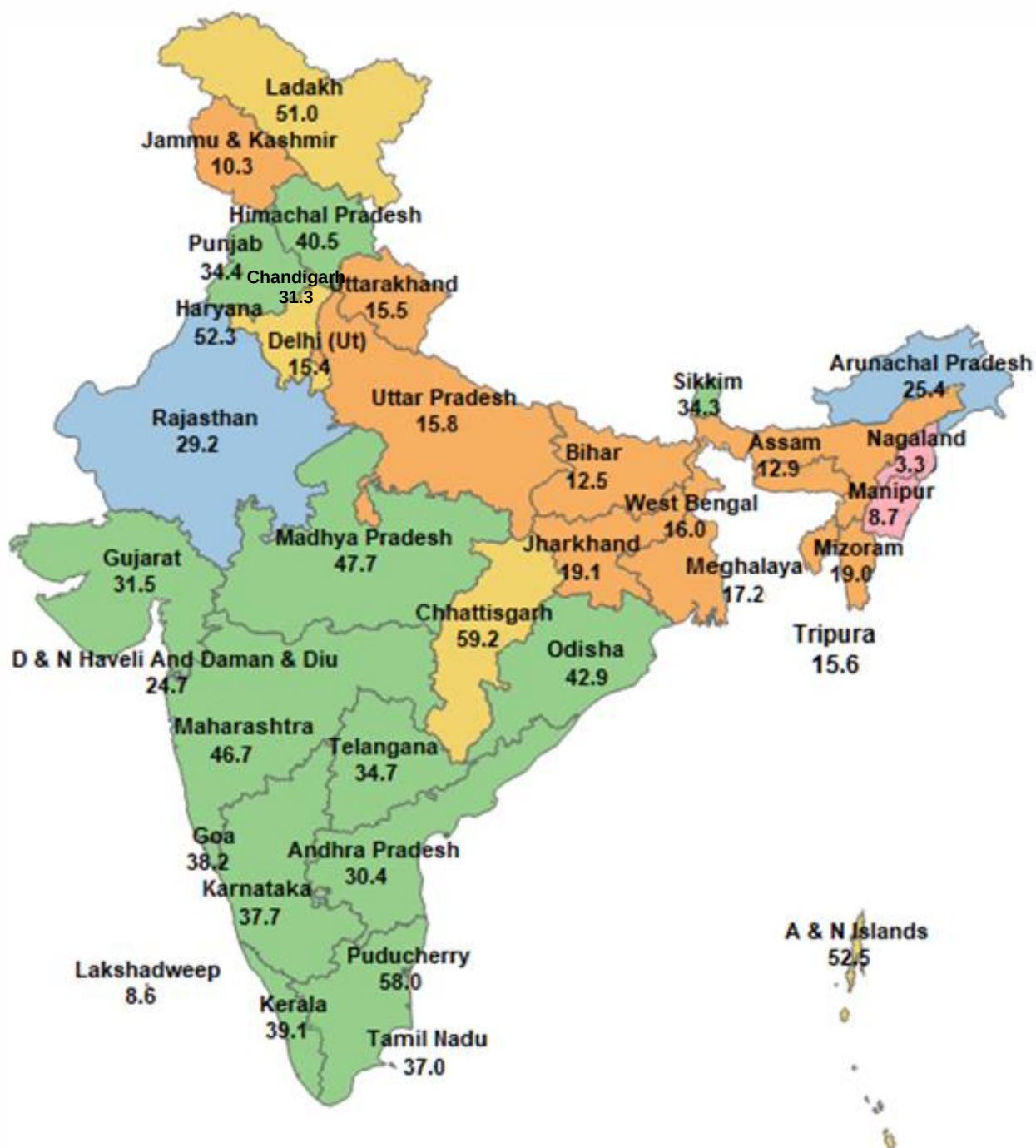
- As per data provided by States/UTs.

Among the States/UTs with significant number of accidental deaths; Tamil Nadu has reported the highest percentage increase (53.8%) in total accidental deaths in the year 2021 as compared to 2020 (from 18,390 in 2020 to 28,286 in 2021), followed by Uttar Pradesh (15.2%), Telangana (10.8%), Andhra Pradesh (9.5%) and Punjab (9.1%).

Delhi has reported a decline of 22.6% in total accidental deaths in the year 2021 as compared to 2020 followed by Chhattisgarh (12.6%) and Assam (7.9%).[**Table-1.3**]

Gender-wise analysis reveals that male and female constitute 81.6% and 18.4% of total victims respectively. Age group of majority of the victims was between 30 to 45 years. This group of people has accounted for 30.9% of all persons killed in accidents in the country during the year 2021. A total of 45,596 senior citizens (60 years & above of age) have died in various accidents during 2021 comprising 11.5%. [**Table-1.4**].

FIGURE-1.6
STATE/UT – WISE RATE OF ACCIDENTAL DEATHS DURING 2021



Note:

Rate of Accidental Deaths means Number of Accidental Deaths per 1,00,000 population.

- As per data provided by States/UTs.

Accidental Death Rate
All India Average 29.1

	upto 10
	10.1 to 20
	20.1 to 30
	30.1 to 50
	Above 50

Causes of Accidental Deaths

Accidental deaths have been grouped into two broad categories i.e. 'Deaths due to Forces of Nature' and 'Deaths due to Other Causes'. Classification of types and causes of accidental deaths has been done for statistical purpose only.

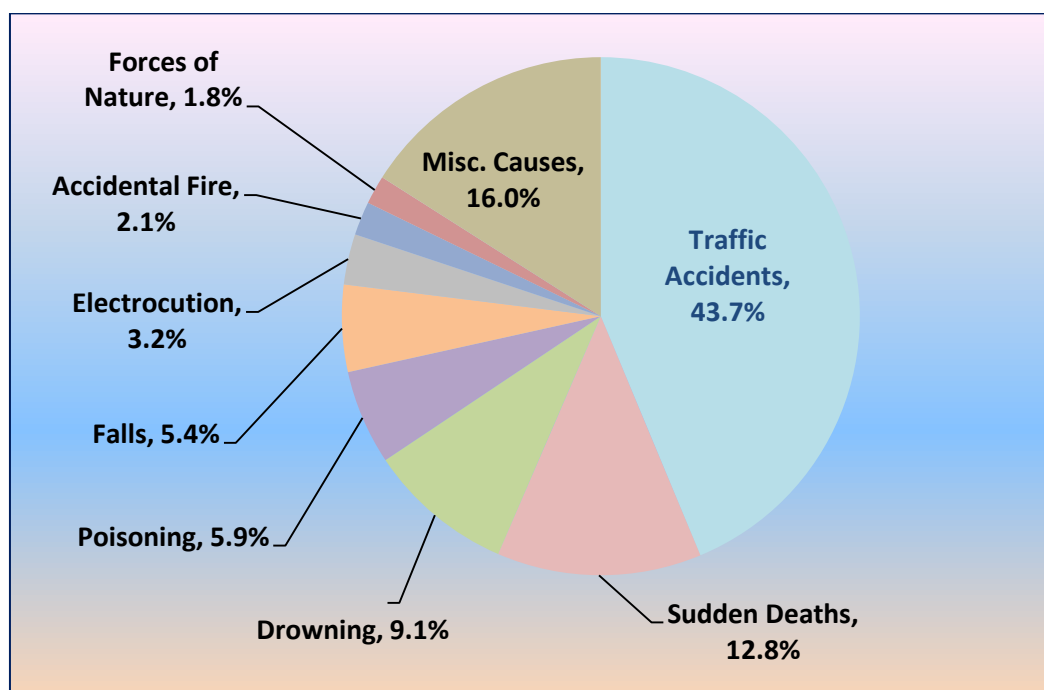
Under causes attributable to 'Forces of Nature', the deaths due to 'Cyclone', 'Tsunami', 'Landslide', 'Lightning', 'Torrential Rain', 'Forest Fire' have increased during the year 2021 over the previous year 2020 whereas the deaths due to *inter-alia* 'Avalanche', 'Exposure to Cold', 'Tornado', 'Flood' and 'Heat/Sun Stroke', have decreased in 2021 over 2020.

The deaths due to 'Collapse of Structure', 'Falls', 'Factory/ Machine Accidents', 'Mines or Quarry Disaster', 'Traffic Accidents', 'Stampede', 'Sudden Deaths', 'Poisoning' and 'Drug overdose' have increased whereas the deaths due to Other Causes such as 'Air Crash', 'Drowning', 'Electrocution', 'Accidental Explosion', 'Accidental Fire', 'Firearms', 'Deaths of Women during Pregnancy', 'Killed by Animals', 'Deaths due to Consumption of Illicit/Spurious Liquor' and 'Suffocation' have decreased in 2021 as compared to the previous year 2020.

There has been an increase in number of traffic accidents deaths during 2021 (1,73,860 deaths) over during 2020 (1,46,354 deaths).

Cause-wise details of accidental deaths, its share and percentage variation during 2021 over 2020 are presented in **Table-1.0**.

FIGURE-1.7
Percentage Share of Various Major Causes of Accidental Deaths during 2021
(Forces of Nature and Other Causes)



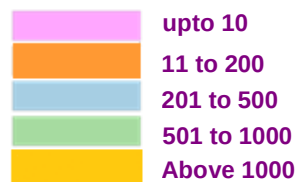
- As per data provided by States/UTs.

Misc. Causes include Collapse of Structure, Suffocation, Deaths of Women during Pregnancy, Killed by Animals, Deaths due to Consumption of Illicit/Spurious Liquor, Accidental Explosion, Drug Overdose, Factory/Machine Accidents, Firearm, Mines or Quarry Disaster, Stampede, Air-Crash, Ship Accidents, Other than above & Unknown Causes.

FIGURE-1.8
STATE/UT – WISE DEATHS DUE TO FORCES OF NATURE DURING 2021



Accidental Deaths



- As per data provided by States/UTs.

Accidental Deaths due to Forces of Nature

Cause - wise accidental deaths due to forces of nature are presented in **Table-1.0**. Out of 7,126 accidental deaths due to forces of nature, 40.4% deaths due to 'Lightning' (2,880 deaths), 9.2% deaths due to 'Flood' (656 deaths), 8.7% deaths due to 'Exposure to Cold' (618 deaths) and 5.3% deaths due to 'Landslide' (380 deaths) during the year 2021. 'Heat/Sun Stroke' and 'Cyclone' caused 5.2% (374 deaths) and 1.7% (118 deaths) of total deaths due to forces of nature respectively.

The age group-wise distribution of accidental deaths due to forces of nature is given in **Table-1.5**. As per age-wise classification of deaths due to forces of nature, 7.2% of such victims were below 14 years of age (514 out of 7,126 deaths), 5.1% in the age-group of 14 years to below 18 years (362 deaths), 16.7% in the age-group of 18 years to below 30 years (1,187 deaths), 24.0% in the age-group of 30 years to below 45 years (1,708 deaths) and 20.2% victims were in age group 45 years to below 60 years (1,437 deaths) during 2021. Senior citizens (60 yrs. and above) constituted 26.9% (1,918 deaths) of the total such victims.

State/UT - wise and gender - wise distribution of accidental deaths in the country by forces of nature along with their share in total deaths is presented **Table-1.9**. 'Lightning' was the dominant factor and caused 94.3%, 83.8%, 80.5% and 78.6%, of total such deaths in Chhattisgarh (230 out of 244 deaths), Tamil Nadu (109 out of 130), Rajasthan (132 out of 164 deaths) and Karnataka (121 out of 154 deaths) respectively. 60.0% (27 out of 45 deaths) of deaths in Kerala was due to 'Flood'. 58.9% (149 out of 253 deaths) of death in Punjab were due to 'Exposure to Cold'. 'Heat/Sun Stroke' was the dominant factor and caused 38.4% (43 out of 112 deaths) of total accidental deaths due to forces of nature in Telangana. 'Landslide' was the dominant factor and caused 100.0%, 75.0%, 61.2% and 50.0% of total deaths in Arunachal Pradesh (12 out of 12 deaths), Sikkim (6 out of 8 deaths), Himachal Pradesh (82 out of 134 deaths) and Mizoram (1 out of 2 deaths) respectively during 2021.

Majority of accidental deaths due to 'Cyclone' were reported in Maharashtra which accounted for 61.0% (72 out of 118 deaths) of total such deaths reported in the country during 2021. Majority of deaths due to 'Torrential Rain' were reported in Uttarakhand which accounted for 25.4% (16 out of total 63 deaths) of total such deaths in the country during 2021.

Accidental Deaths due to Other Causes

It is generally accepted that accidents due to 'Other Causes' are preventive and can be reduced by effective safety measures and safety consciousness.

A total of 6,38,691 cases of Accidents due to 'Other Causes' were reported in the country which resulted in 3,90,404 deaths and 3,77,070 persons injured during 2021 [**Table-1.7**].

The cause-wise analysis of deaths revealed that majority of deaths were due to traffic accidents which accounted for 44.5% (1,73,860 out of 3,90,404 total deaths) of total deaths due to 'Other Causes' during the year 2021. Besides, 'Sudden Deaths', 'Drowning', 'Poisoning' and 'Falls' caused 13.0% (50,773 deaths), 9.3% (36,362 deaths), 6.0% (23,472 deaths) and 5.5% (21,609 deaths) of total deaths due to other causes respectively. [**Table-1.0**]

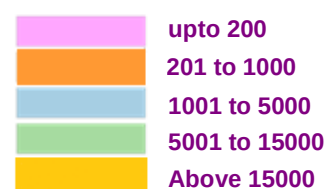
The age-group-wise distribution of accidental deaths in the country by various 'Other Causes' along with their share in total deaths is presented in **Table-1.8**. Majority of victims were in age groups 30 - below 45 years and 18 - below 30 years which accounted for 31.1% (1,21,235 deaths) and 26.1% (1,01,778 deaths) of total such deaths respectively.

State/UT-wise and gender-wise distribution of accidental deaths in the country by various other causes is presented in **Table-1.10**. Maharashtra with 9.1% of country's population has reported 14.7% (57,481 out of 3,90,404 deaths) of accidental deaths due to other causes. Madhya Pradesh, Uttar Pradesh, Tamil Nadu, Karnataka, Rajasthan and Gujarat accounted for 10.2%, 9.3%, 7.2%, 6.4%, 5.9% and 5.6% of total such deaths respectively during the year 2021.

FIGURE-1.9
STATE/UT – WISE ACCIDENTAL DEATHS DUE TO OTHER CAUSES
DURING 2021



Accidental Deaths



- As per data provided by States/UTs.

State/UT-wise and Cause-wise details on 'Accidents due to Other Cause' is presented in **Table-1.10**. State/UT-wise analysis reveals that Uttar Pradesh accounted for maximum deaths (24,711 out of 1,73,860) under traffic accidents. Maharashtra accounted for maximum deaths (14,927 out of 50,773) under 'Sudden Deaths'. Maharashtra accounted for maximum deaths (5,056 out of 36,362) under 'Drowning'. Madhya Pradesh accounted for maximum deaths (5,351 out of 23,472) under 'Poisoning'.

A total of 708 incidents of consumption of illicit/spurious liquor caused 782 deaths in the country during 2021. States where various such deaths were reported include Uttar Pradesh (137 deaths) followed by Punjab (127 deaths), Madhya Pradesh (108 deaths), Karnataka (104 deaths), Jharkhand (60 deaths) and Rajasthan (51 deaths).

Fire Accidents

Considering the number of fire accidents in the country and potential of the damage that can be caused by such accidents, the Bureau has also collected data on fire accidents according to places of occurrence for appropriate preventive strategies viz. 'Fire in School Buildings', 'Fire in Commercial Buildings', 'Fire in Residential / Dwelling

Buildings', 'Fire in Government Buildings', 'Fire in Mines', 'Fire in Factories', 'Accidental Fire in Trains' and 'Accidental Fire in Vehicles' [**Table-1.11**].

A total of 8,491 cases of fire accidents were reported in the country during 2021, showing a decrease of 9.0% as compared to 2020 (9,329 cases). 8,491 incidents of fire accidents caused injuries to 485 persons and 8,348 deaths during 2021. The cause-wise analysis of fire accidents revealed that 50.8% of total deaths (4,240 out of 8,348) were reported in residential/dwelling buildings during 2021 [**Table 1.12**].

Large number of cases of fire accidents were reported in Odisha (1,248 out of 8,491), accounting for 14.7% of total such cases during 2021 [**Table-1.12**].

The State/UT-wise analysis of place of fire accidents revealed that 22 States & UTs have reported 50.0% or more deaths due to 'Fire in residential or dwelling building' during 2021. These States/UTs were Mizoram (100.0%), Assam (95.7%), Himachal Pradesh (88.2%), Karnataka (86.9%) and Tripura (86.4%). [**Table-1.12**].

Accidents in Cities

The term 'Mega City' refers to a city which has a population of 10 lakhs or more. As per the Population Census 2011, the status of mega city has been accorded to 53 cities.

A total of 53,497 accidental deaths were reported in 53 mega cities during 2021. Maximum numbers of accidental deaths (6,905 deaths) were reported in Mumbai accounting for 12.9% of total deaths, followed by Bengaluru (3,613), Delhi City (3,074), Pune (2,900), Surat (2,288) and Jaipur (2,117).

Rate of accidental deaths in these 53 mega cities was higher (33.3) than all India rate of accidental deaths (29.1). The rate of accidental deaths was highest in Rajkot (92.9), followed by Faridabad (90.9), Raipur (89.0), Asansol (86.7), Ludhiana (77.2) and Nagpur (76.6) as compared to average rate for the cities (33.3) [Table-1.2].

Out of 53,497 accidental deaths in 53 mega cities; 533 and 52,964 deaths were due to 'Forces of Nature' and 'Other Causes' respectively during 2021. An increase of 71.4% was observed in deaths due to forces of nature and an increase of 5.5% was witnessed in deaths due to 'Other Causes' during 2021 over 2020 respectively. [Table-1.3]

Kolkata, the third largest city in terms of population (8.8% of the total population of all mega cities) among 53 cities, reported the low share of deaths in cities accounting for 0.6% of total accidental deaths in 53 mega cities during 2021. The comparative reporting of 'Accidental Deaths' during 2021 in the top four-metropolitan cities is presented in LIST-1.3.

Accidental Deaths due to Forces of Nature in Cities

A total of 533 deaths due to forces of nature were reported in these 53 mega cities during 2021. Among specified causes of accidents due to forces of nature, maximum deaths were reported due to 'Exposure to cold' followed by 'Cyclone', accounting for 20.1% and 13.3% of total such accidental deaths. Due to 'Exposure to cold' maximum deaths were reported in Ludhiana (39 deaths), Amritsar (21 deaths) and Dhanbad (15 deaths). Due to 'Cyclone' deaths were reported in Mumbai (71 out of 71 deaths).

Accidental Deaths due to Other Causes in Cities

A total of 95,362 cases of 'Other' accidents were reported in 53 mega cities which caused injuries to 48,220 persons and 52,964 deaths during 2021. The cause-wise analysis of deaths due to other causes revealed that 27.8%, 26.3%, and 7.5% deaths in 53 cities were due to "Traffic Accidents" (14,714), Sudden deaths' (13,955), and 'Falls' (3,964) respectively.

Among 53 cities, Mumbai city (6,716) accounted for 12.7% of total deaths reported under 'Other' accidents followed by Bengaluru (3,613) and Delhi city (3,074) which accounted for 6.8% and 5.8% respectively during 2021.

"Data on City-wise information is available on our website under 'Additional Tables for ADSI'"

LIST-1.3
Rate of Accidental Deaths in Metropolitan Cities during 2021

Sl. No.	City	Population (In Lakhs)	Share of Population In Cities (Percent)	Share of Deaths In Cities (Percent)	Rate (No. of Deaths Per One Lakh Population)
(1)	(2)	(3)	(4)	(5)	(6)
1.	Chennai	87.0	5.4	3.2	19.6
2.	Delhi	163.2	10.2	5.7	18.8
3.	Kolkata	141.1	8.8	0.6	2.2
4.	Mumbai	184.1	11.5	12.9	37.5

- As per data provided by States/UTs.
